

CS 4720 HW10

1) Over here, we have explored the following squares:

- $[1,1]$ has no breeze, so $b_{[1,1]}$ becomes 0
- $[1,2]$ has breeze, so $b_{[1,2]}$ becomes 1
- $[2,1]$ has breeze, so $b_{[2,1]}$ becomes 1

And we have the following knowledge:

$$\text{known} = \neg p_{1,1} \wedge \neg p_{1,2} \wedge \neg p_{2,1}$$

We have the prior knowledge as well:

$$P(p_{i,j}) = 0.01, \text{ independent squares}$$

We have the following unknown neighbors for breezy squares:

- $[1,2] = \{[1,2], [2,2], [1,1]\}$
- $[2,1] = \{[2,2], [2,0], [3,1], [1,1]\}$ (However, only $[2,2]$ and $[3,1]$ are valid squares, so that's why they matter)

Let us consider the following variables to represent $p_{[1,1]}$, $p_{[2,2]}$ and $p_{[3,1]}$:

$$X = p_{[1,1]}$$

$$Y = p_{[2,2]}$$

$$Z = p_{[3,1]}$$

Now, let us consider the breeze constraints for $[1,2]$. We have $b_{[1,2]}$ as 1, implies that $X \vee Y$ is also 1 and for $[2,1]$ we have $b_{[2,1]}$ as 1, which implies that $Y \vee Z$ is also 1, and also we have $p_{[3,1]} = Z \sim \text{Bernoulli}(0.01)$.

Let us now list all the valid pit configurations. We have variables such as X, Y, Z where each one of them belongs to $\{0, 1\}$ with prior 0.01, plus we have the following constraints from breezes:

- $X \vee Y = 1$
- $Y \vee Z = 1$

Now, let us try and enumerate all the variables (X, Y, Z) that satisfy both:

X	Y	Z	Does it satisfy breeze constraints?	Probability
0	0	0	It fails both the constraints	-
0	0	1	The b(1,2) fails in this case	-
0	1	0	Yes, it satisfies both of them	$0.99 \times 0.01 \times 0.99 = 0.009801$
0	1	1	Yes, it satisfies both of them	$0.99 \times 0.01 \times 0.01 = 0.000099$
1	0	0	b(2,1) fails in this case	-
1	0	1	b(2,1) fails in this case	-
1	1	0	Yes, it satisfies both of them	$0.01 \times 0.01 \times 0.99 = 0.000099$
1	1	1	Yes, it satisfies both of them	$0.01 \times 0.01 \times 0.01 = 0.000001$

Now, we can say that the total probability mass can be found out summing all the probabilities for the fields that tell us that the value of the variables satisfies the given breeze constraints:

$$Z = 0.009801 + 0.000099 + 0.000099 + 0.000001 = 0.01$$

Now, let us compute $P(P_{1,3} = 1 \mid \text{known}, b)$. Over here, we know that $P(X=1)$ is the sum of probabilities with $X = 1$, so we have $(1,1,0) = 0.000099$, $(1,1,1) = 0.000001$ and $P(1,0,*)$ is invalid, so we can calculate $P(P_{1,3} = 1 \mid \text{known}, b)$ as follows:

$$P(X = 1 \mid b) = 0.0001/0.01 = 0.01 \quad (0.000099 + 0.000001 = 0.0001)$$

Now, let us compute $P(P_{2,2}=1 \mid \text{known}, b)$ We have these with $Y = 1$:

- $(0,1,0) = 0.009801$
- $(0,1,1) = 0.000099$
- $(1,1,0) = 0.000099$
- $(1,1,1) = 0.000001$

$$\text{Sum} = 0.009801 + 0.000099 + 0.000099 + 0.000001 = 0.01$$

$$P(Y=1 \mid b) = 0.01/0.01 = 1$$

Hence, we our answers our $P(P_{1,3} = 1 \mid \text{known}, b) = 0.01$ and $P(P_{2,2}=1 \mid \text{known}, b) = 1$.

2a) Network c, because it does not have any arrows between the parent nodes.

b) Networks a and b make because they show that a child's genes basically depends on their parent's genes and the child's handedness depends on child himself.

c) Since network b adds extra dependencies by stating that the child's handedness directly influences the child's handedness and c says that the genes are independent because everyone's gene's are assigned independently at birth, with no inheritance and handedness comes from own's genes but they are not related across any individuals. So, basically network a is the only one that says handedness is determined by a gene and the gene is inherited from the parents which is what the original statement says as well.

d) The model says that over here each here each person has a gene and basically over here the child's gene is basically inherited from either parent with a probability of 0.5 each. then mutate with a probability m , so we can say that the process for G_{child} must start with randomly picking up mother and father, copy the parent's gene with probability m , flip $l \longleftrightarrow r$, with a probability of $1-m$ and keep it. Next, we must fill in the values for $P(G_{\text{child}}=l \mid G_{\text{mother}}, G_{\text{father}})$ and $P(G_{\text{child}}=r \mid G_{\text{mother}}, G_{\text{father}})$ where for all 4 combinations of $(G_{\text{mother}}, G_{\text{father}})$ belong to $\{l, r\}^2$. So, we have 2 cases over here:

1) Case 1

$$G_{\text{mother}} = l, G_{\text{father}} = l$$

In this case child picks a parent, mom(l) with prob 0.5 and dad(l) with prob 0.5. Over here, in both the cases the initial gene is l so it stays l with prob $1-m$ and mutates to r with prob m , so overall.

$$\text{So, now we have } P(G_c = l \mid l, l) = 1 - m, P(G_c = r \mid l, l) = m$$

$$\text{Case 2: } G_{\text{mother}} = l, G_{\text{father}} = r$$

So we have 2 ways to do it:

We have 2 ways to do it:

- It basically inherits from mother (l) with prob 0.5 which stays l with prob $1-m$ and it becomes r with prob m .

- It must inherit from father(r) with prob 0.5 stays r with prob 1-m and becomes l with prob m.

Now, let us compute $P(G_c = l \mid l, r)$:

$$P(G_c = l \mid l, r) = 0.5 * P(\text{child gets l via mom}) + P(\text{child gets l via dad}) = 0.5 [(1-m) + [0.5]] = 0.5$$

So, now we have $P(G_c = l \mid l, r) = 0.5$, $P(G_c = r \mid l, r) = 0.5$

Case 3: Mother r, father l

$$G_{\text{mother}} = r \text{ and } G_{\text{father}} = l$$

By looking at the logic of case 2, we can say that $P(G_c=l \mid r, l)=0.5, P(G_c=r \mid r, l)=0.5$

Case 4: If both parents have gene r

$$G_{\text{mother}} = r \text{ and } G_{\text{father}} = r$$

Now, both parents basically start as r, similar to case we will have the following logic:

- Initial gene always r
- It basically stays with r with prob 1-m, mutates to l with prob m, so now we have:

So, we get:

$$P(G_c = l \mid r, r) = m, \quad P(G_c = r \mid r, r) = 1-m.$$

Hence, we can display the CPT for each child in the following way:

G_{mother}	G_{father}	$P(G_{\text{child}}=l)$	$P(G_{\text{child}} = r)$
l	l	1-m	m
l	r	0.5	0.5
r	l	0.5	0.5
r	r	m	1-m

e) We have been given that $P(G_{\text{father}}=l)=P(G_{\text{mother}}=l)=q$

Now, let us assume the parents' genes are independent in the prior, so:

Let us assume that the parents' genes are independent of each other in the prior, so $P(G_{\text{mother}}, G_{\text{father}}) = P(G_{\text{mother}})P(G_{\text{father}})$ and we want $P(G_{\text{child}} = l)$.

Now, let us look at the condition on all combinations of parents' genes. By using the law of total probability we should list all the 4 cases:

$$P(G_c=l) = P(G_c=l \mid l,l)P(l)P(l) + P(G_c=l \mid l,r)P(l)P(r) + P(G_c=l \mid r,l)P(r)P(l) + P(G_c=l \mid r,r)P(r)P(r)$$

Now, let us plugin all the values from d, so we get

$$P(G_c=l) = (1-m)q \cdot q + 0.5q \cdot (1-q) + 0.5(1-q) \cdot q + m(1-q) \cdot (1-q)$$

Now, if we add the middle 2 terms we get $0.5q(1-q) + 0.5q(1-q) = q(1-q)$, so now we have $P(G_c=l) = (1-m)q^2 + q(1-q) + m(1-q)^2$.

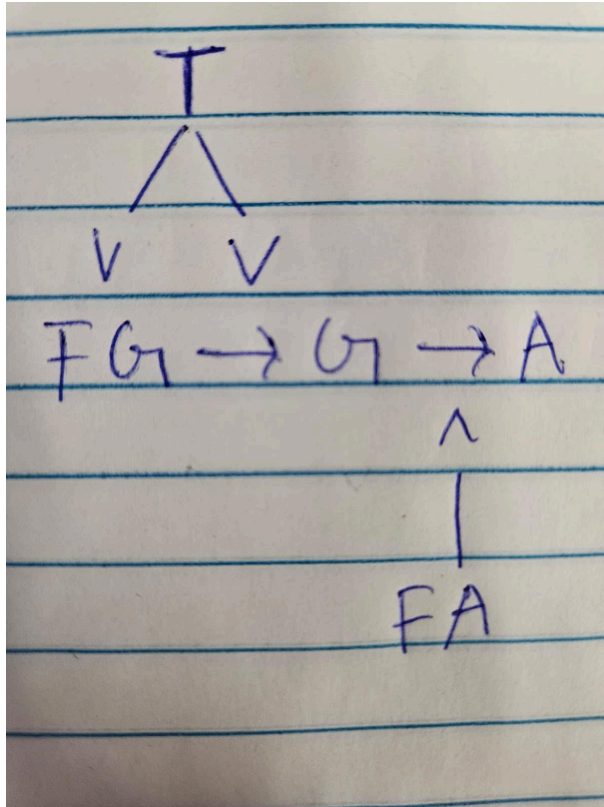
Now, let us expand $q(1-q)$ and $(1-q)^2$, so we get $q(1-q) = q - q^2$ and $(1-q)^2 = 1 - 2q + q^2$. So, after substituting we get and doing the calculation:

$$P(G_c=l) = m + q(1-2m)$$

Hence, our final expression is $P(G_c=l) = m + q(1-2m)$.

3a) We have the following arrows over here:

$T \rightarrow FG$
 $T \rightarrow G$
 $FG \rightarrow G$
 $G \rightarrow A$
 $FA \rightarrow A$



This network basically encodes that high temperature increases the probability of gauge failure ($T \rightarrow FG$), the gauge reading depends on both the true temperature and whether the gauge is faulty and also the alarm depends on gauge reading and whether the alarm is faulty.

c) Over here, we have been given that there are two possible actual temperatures (normal and high), two possible gauge readings (normal or high), if gauge is working ($FG = \text{false}$) $\rightarrow$ probability gauge is correct = x , if gauge is faulty ($FG = \text{true}$) $\rightarrow$ probability gauge is correct = y

Now, we already know that "Correct" means $G = T$ and "Incorrect" means G is not equal to T , so we must give $P(G=g \mid T=t, FG=f)$ for all the combinations of T and FG .

Now, we can calculate the value of CPT is can be calculated using the following table:

T(actual)	FG(faulty)	P(G=actual)	P(G is not equal)
Normal	false	x	$1-x$
high	false	x	$1-x$
normal	true	y	$1-y$
high	true	y	$1-y$